

Agent-Based Modeling of a Zero-Emission Zone in Berlin Using MATSim: Fleet Electrification, Adaptation Dynamics, and Assessment of Emission Impacts

Jaroslav Smirnov, Prof. Dr. Kai Nagel, Tim Volotskiy, Tina Rakic, Dr.-Ing. Tilmann Schlenther, Ricardo Ewert

Keywords: MATSim, Zero-Emission Zone, Fleet Electrification, Travel Behavior

Context and Research Question

Berlin, like many major European cities, aims to substantially reduce transport-related emissions in order to achieve its climate targets. One of the measures under discussion is the introduction of a Zero-Emission Zone (ZEZ), restricting access for vehicles with internal combustion engines in selected areas of the city.

This study investigates a scenario in which a ZEZ is introduced within Berlin's S-Bahn Ring by 2035. The research addresses how such a measure affects travel behavior, trip patterns, the level of fleet electrification, and transport-related emissions compared with a scenario without a ZEZ that nevertheless accounts for the ongoing transition toward electric vehicles.

The analysis is based on the agent-based transport simulation framework MATSim. In contrast to aggregate transport models, MATSim allows changes in individual travel behavior to be analyzed, including route choice, mode choice, and vehicle technology choice. This makes it possible to quantitatively assess different mechanisms of adaptation to a ZEZ, such as switching to electric vehicles, shifting to public transport or cycling, and changing travel patterns. Furthermore, MATSim enables the impacts of the policy on different socio-economic population groups to be investigated using the built-in MATSim scoring framework.

MATSim Model and Scenario Structure

The simulations are based on the open MATSim model of the Berlin metropolitan region, version 6.4. The model's base year is 2023, while the public transport schedule has been updated to represent the 2024 service level.

Agents are represented by activity-based daily plans and can choose between the transport modes `car`, `ride`, `public transport`, `bike`, and `walk`. For comparison with the German Mobility in Germany (MiD) survey, `car` and `ride` are combined into the category of motorized individual transport (MIT). In scenarios incorporating fleet electrification, the additional modes `ev_car` and `ev_ride` are introduced. Commercial and freight transport are represented by separate agent categories.

Simulated Scenarios

Three main scenarios are considered.

The **baseline scenario** represents the transport system without additional access restrictions.

The **EV-relaxed** scenario allows agents to voluntarily switch to electric vehicles. EV use is represented by higher fixed daily costs combined with lower variable operating costs per kilometer.

The **EV&ZEZ-relaxed** scenario additionally introduces a Zero-Emission Zone. All road links located within the S-Bahn Ring become inaccessible to vehicles with internal combustion engines, while electric vehicles of all categories retain unrestricted access.

For the 2035 forecast, the results of the **EV-relaxed** and **EV&ZEZ-relaxed** scenarios are further transformed using a fleet diffusion model, allowing the gradual penetration of electric vehicles into the vehicle fleet to be represented.

Modeling the ZEZ in MATSim and Equilibrium Considerations

The ZEZ is implemented directly at the road-network level. All road links located within the area enclosed by the S-Bahn Ring, including an additional buffer of approximately 500 m, become inaccessible to vehicles with internal combustion engines. For the `car` mode, these links are removed from the accessible network, whereas electric vehicles retain access.

User equilibrium is approached using the `SubtourModeChoice` algorithm, extended to include the `ev_car` and `ev_ride` modes. During the simulation of voluntary EV adoption, a convergence problem was identified due to the relatively high fixed daily cost of vehicle use. Since most agents in the initial population started with plans using the `car` mode, transitions to `ev_car` occurred only rarely. This resulted in a local equilibrium with a substantially underestimated share of electric vehicles.

To address this issue, an additional plan was introduced for each relevant agent in the initial population. This plan was identical to the original plan, except that all `car` trips were replaced by `ev_car` trips. The initial score of these plans was set to `null`, allowing MATSim to evaluate their utility during the first iterations. Providing this alternative initial plan substantially improved exploration of the solution space and helped avoid convergence toward a local equilibrium. In the scenarios considered here, this increased the final share of agents using electric vehicles by approximately 25%.

Fleet Electrification and Diffusion Model

MATSim determines a long-term behavioral equilibrium under a given set of conditions, whereas actual vehicle fleet renewal takes place gradually over time. Therefore, for the 2035 forecast, the MATSim results were complemented by a separate EV diffusion model.

For the scenario without a ZEZ, the parameters of the diffusion curve were calibrated using statistics on new electric vehicle registrations in Berlin. The EV share obtained in the **EV-relaxed** scenario was used as the target saturation level, while the average vehicle lifetime was assumed to be 13 years.

For the **EV&ZEZ** scenario, two independent diffusion curves were applied: one for residents within the ZEZ and another for residents outside the zone. The introduction of the ZEZ was assumed to substantially accelerate fleet renewal among residents of the restricted area because regular use of internal combustion engine vehicles would no longer be possible. The average fleet renewal period within the ZEZ was assumed to be 6.5 years. This represents both accelerated vehicle replacement by owners and the redistribution of vehicles through the second-hand vehicle market. For residents outside the ZEZ, the parameters of the baseline diffusion process were retained.

The final EV share in 2035 was determined by aggregating the two trajectories using weights corresponding to the trip volumes generated by each population group. This approach combines the long-term behavioral equilibrium obtained from MATSim with a more realistic representation of temporal fleet renewal dynamics.

Emission Assessment

Transport emissions were estimated using the MATSim `emissions contrib` module. The calculation was performed by post-processing the MATSim events file. For each road-link traversal, the link length, entry and exit times, and resulting average travel speed were determined. Based on these parameters, the corresponding emission factor was selected from the HBEFA database.

Both operating emissions during vehicle movement (`warm emissions`) and emissions associated with cold engine starts (`cold-start emissions`) were included. Electric vehicles were represented as `NON_HBEFA_VEHICLE` and therefore produced no direct exhaust emissions.

Main Transport Simulation Results

In the **EV-relaxed** scenario, lower variable operating costs of electric vehicles lead to a moderate increase in motorized individual transport. The MIT mode share increases from 23.84% in the baseline scenario to 24.68%, indicating a rebound effect. At the same time, the share of electric MIT reaches 13.35%, while the share of trips made by internal combustion engine vehicles decreases to 11.33%.

The introduction of the ZEZ changes the structure of transport demand. In the **EV&ZEZ-relaxed** scenario, the MIT share among Berlin residents decreases to 22.02%, which is 2.66 percentage points lower than in the **EV-relaxed** scenario. The strongest changes are observed among residents of the ZEZ: their MIT share decreases from 16.05% to 11.21%, while the use of internal combustion engine vehicles almost disappears, reaching only 0.27%. Electric vehicle trips account for 10.95%. At the same time, public transport use increases to 36.61% and cycling to 23.60%.

After accounting for the fleet electrification diffusion process, the main trends remain unchanged. In the **EV'35** scenario, MIT accounts for approximately 25% of trips, with around 9% made by electric vehicles and approximately 15% by internal combustion engine vehicles. In the **EV&ZEZ'35** scenario, the MIT share decreases to approximately 22%, while the electric MIT share increases to around 13% and the share of internal combustion engine vehicles falls to approximately 9%. Among ZEZ residents, almost all remaining car traffic is served by electric vehicles.

A comparison between the **EV&ZEZ'35** and **EV'35** scenarios indicates an additional reduction in direct CO₂ emissions of approximately **2,413 t/day**. Passenger cars account for approximately **1,033 t/day** of this reduction, light commercial vehicles (LCVs) for approximately **855 t/day**, and trucks for approximately **524 t/day**. Additional reductions are observed for other pollutants, including **5.89 t/day of NO_x**, **19.72 t/day of CO**, **0.90 t/day of HC**, and **0.07 t/day of PM**.

The results indicate that introducing a Zero-Emission Zone can not only accelerate fleet electrification but also reduce individual motorized transport in the central urban area by shifting trips toward public transport and cycling. The combination of agent-based transport simulation and fleet diffusion modeling makes it possible to capture both behavioral adaptation and the gradual technological transition of the vehicle fleet, providing a comprehensive framework for assessing the transport and emission impacts of urban zero-emission policies.